Summary in Graph

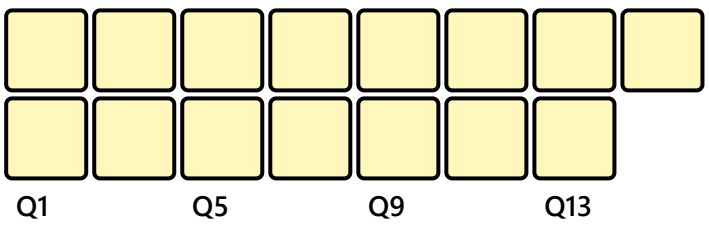
Exam Summary (GO Classes GATE CS | Computer Networks | Weekly Quiz 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (8 + 7)
Total Marks:	22 (8 + 14)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (31) out of 309 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

In a Go-Back-N (GBN) protocol, the sender uses a window size of 4 and sequence numbers range from 0 to 1023 (modulo 1024). Suppose the sender had transmitted packets with sequence numbers $500, 501, 502,$ and 503 , and 500 was the base of the sender's window. After a while, the sender receives an ACK for packet 500.

Assume:

- The ACK is cumulative.
- No other ACKs have been received.
- No timeouts have occurred yet.

Which of the following statements is/are correct?

A. The sender can now send packet 504 immediately.
B. The sender must retransmit packet 500, since no ACK for 501 has been received.
C. The new sender window is $[501, 504]$, and packets $501 - 503$ can be retransmitted if timeout occurs.
D. The base of the sender window remains unchanged until ACKs for all packets $500 - 503$ are received.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 01sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A sender using the Go-Back-N protocol has a current window of size 4 , which includes the sequence numbers $100, 101, 102$, and 103 . The sender has sent packets 100 and 101 so far.

Which of the following statements **must be true**?

A. Packet 100 has been sent and acknowledged.
B. Packet 103 is outside the window and cannot be sent yet.
C. If packet 102 is sent, then packet 101 must have already been sent.
D. If packet 102 has not been sent, then packet 101 must not have been sent either.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Two hosts A and B are 2000 km apart and are connected using a direct link. The propagation speed on the link is 2.5×10^8 m/s. Host A is sending packets of size $\mathbf{100}$ bytes to host B over this link. If the transmission rate of the link is R bits per second, how much time does it take to transmit one packet?

A. $\frac{100}{R}$ seconds

B. $\frac{100 \times 8}{R}$ seconds

C. $\frac{2000 \times 10^3}{2.5 \times 10^8}$ seconds

D. $\frac{R}{100 \times 8}$ seconds

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

In a sliding window protocol, the sender and receiver both use a window size of $N = 5$. The sequence number space ranges from 0 to 15 . At a certain time, the sender's window includes the sequence numbers 0 to 4 , and the receiver's window includes the sequence numbers 2 to 6 . Which of the following best describes

the current state?

A. The sender has received ACKs for packets 0 and 1 , and has sent packets 2 to 4.

B. The receiver is expecting packet 0 , as it has not received any packets yet.

C. The sender has sent packets 0 to 4 , and packets 0 and 1 have been acknowledged.

D. The receiver has received packets 0 to 2 , and is waiting for 3 to 7 .

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Suppose it takes 1 ms to send a packet, and the one-way propagation delay between the sender and receiver is $\mathbf{10\ ms}$. The sliding window size is again $N = 4$. What is the channel (or link) utilization?

A. 19.05%

B. 25%

C. 33.3%

D. 40%

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A sender is using a sliding window protocol with a window size of 4 . It has sent packets 0 to 3 , and ACKs for packets $0, 1, 2,$ and 3 are currently propagating from the receiver to the sender. No ACK has yet been received by the sender. Assume cumulative ACKs are used and that the sender takes action only when an ACK is received or a timeout occurs.
Which of the following statements about the next possible event at the sender is/are correct?

A. If the next event is ACK1 (ACK0 was lost), the sender will slide the window and send packet 4.
B. If the next event is ACK2 (ACK0 and ACK1 were lost), the sender will note that packet 2 is ACKed but will not move the window or send any packet.
C. If the next event is ACK0, the sender will slide the window and send packet 4.
D. If all ACKs (ACK0, ACK1, ACK2, and ACK3) are lost, the sender will receive a duplicate ACK for packet 3 and move the window accordingly.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Assume a sender using Go-Back-N ARQ transmits segments 10 to 19 after all previous segments have been acknowledged. The receiver receives segments: $10, 11, 12, 13, 16, 17, 18, 19$. Which ACK will be sent by the receiver?

A. Cumulative ACK till segment 13

B. Cumulative ACK till segment 14
C. ACKs for 13 and 16 separately
D. Cumulative ACK till 10

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

In a Go-Back-N ARQ protocol with a maximum sender window size of 10 segments, what is the minimum number of bits required to represent the sequence number field?

A. 3
B. 4
C. 5
D. 10

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

In a Go-Back-N (GBN) protocol, the sender uses a window size of 4, and sequence numbers range from 0 to 1023 (modulo 1024). At a certain moment, the receiver is waiting for packet number 200 to arrive in order.

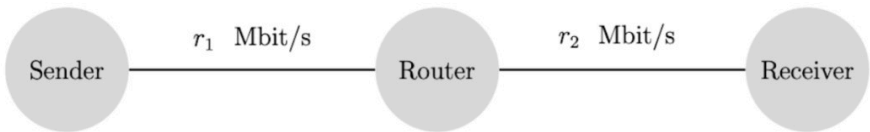
Due to network propagation delay, some ACKs sent earlier by the receiver may still be in transit to the sender. Which of the following ranges correctly describes the possible ACK numbers that the sender may still receive at this point?

A. 200 to 203
B. 197 to 199
C. 196 to 199
D. 196 to 200

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Two packets are sent back-to-back from a sender through a store-and-forward router (i.e., it forwards the packet only after receiving it fully). The first packet has size p_1, and the second packet has size p_2. The transmission rates on the first and second links are r_1 and r_2, respectively.



What is the condition on p_2 such that the queueing delay of the second packet at the router is zero?

A. $p_2 \geq \frac{r_2}{r_1} \cdot p_1$

B. $p_2 \leq \frac{r_2}{r_1} \cdot p_1$

C. $p_2 \leq \frac{r_1}{r_2} \cdot p_1$

D. $p_2 = p_1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Node A wants to send a large amount of data to Node C via an intermediate Node B using the **Selective Repeat protocol** .

- Each **bit** transmitted on link $A - B$ has a probability p of getting corrupted
- Each **bit** transmitted on link B-C has a probability q of getting corrupted
- Bit errors occur **independently** across links and across bits

Each data packet is d bits long, and each ACK is a bits long. The ACK is sent end-to-end from C to A (via B $\rightarrow A$)

What is the expected number of transmissions required by A to successfully send one packet and get it acknowledged?

A. $1/X$, where $X = (1-p)^d(1-q)^d(1-q)^a(1-p)^a$
B. $1/X$, where $X = (1-p)^d(1-q)^a$
C. $1/X$, where $X = (1-p)^d(1-q)^d(1-p)^a$
D. $1/X$, where $X = (1-p)^d(1-q)^d(1-q)^a$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A packet of size N bits is sent from Host A to Host B via switch C using Hop-by-Hop Stop-and-Wait ARQ. Each link $(\mathrm{A} \rightarrow \mathrm{C}$ and $\mathrm{C} \rightarrow \mathrm{B})$ has the same packet success probability P_{packet} . The transmission rate on both links is R bits per second. Ignoring ACK transmission delays, queueing, and processing delays, what is the expected total transmission delay (in seconds) for a packet to be successfully delivered from A to B ?

A. $\frac{N}{R}$

B. $\frac{2N}{R}$

C. $\frac{2N}{R \cdot P_{\text{packet}}}$

D. $\frac{2N}{R} \cdot \frac{1}{P_{\text{packet}}^2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A Go-Back-N protocol uses a window size of 5 and cumulative acknowledgments. The parameters are as follows:

- Timeout $= 100$ ms, measured from the time the first bit of the packet is sent
- Packet transmission time $= 1$ ms
- One-way link latency $= 10$ ms

The sender begins transmitting packets 0 to 4 at $t = 0$. All packets are received correctly except packet number 3, which is lost.

When is packet 3 first retransmitted (i.e., sent again after timeout expiry)?

A. $t = 24$ ms
B. $t = 102$ ms
C. $t = 103$ ms
D. $t = 100$ ms

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A sender and receiver implement a sliding window protocol using selective retransmission with the following parameters:

- Frame sequence number space: 0 to 4
- Send window and receive window size: 2
- One-way propagation delay: slightly less than 1 ms
- Sender transmits one packet per ms
- Timeout interval at sender: 4 ms
- Receiver sends cumulative ACKs
- ACK 1 is lost (as shown in the figure)

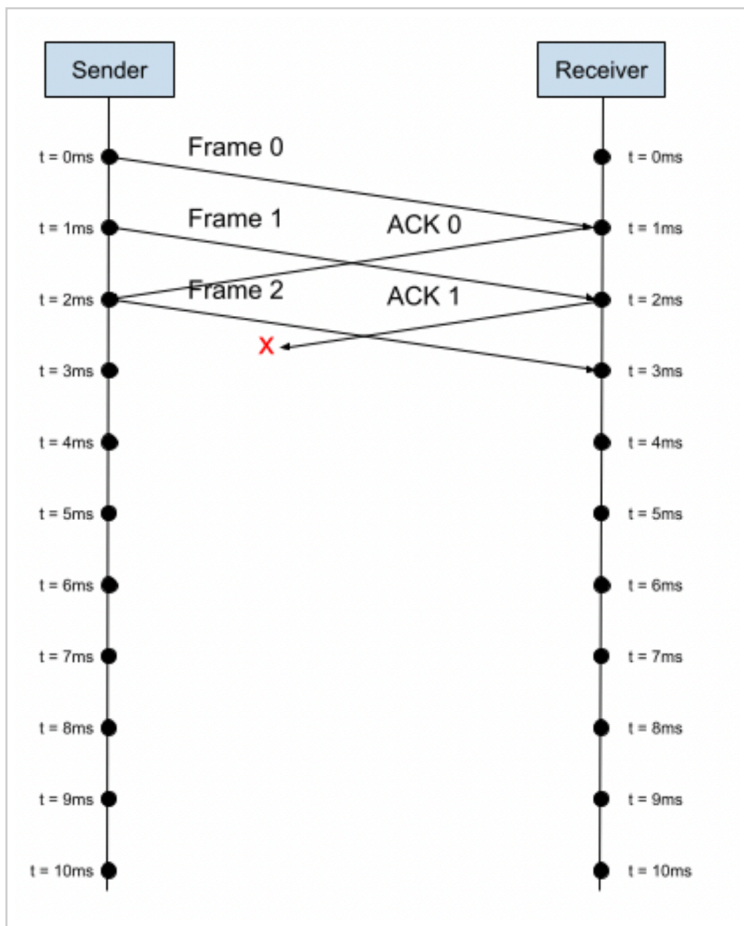


Figure: Sliding Window Timeline

Using the above setup and the timeline of packet exchanges shown, which of the following statements are correct?

> Note: In this question, `ACK n` refers to the cumulative acknowledgment for Frame n, i.e., all frames up to and including Frame n have been received successfully.

A. The sender receives ACK 4 at t = 7 ms, confirming delivery of Frame 4.

B. Frame 0 is retransmitted at t = 6 ms, even though it was previously acknowledged.

C. ACK 1 is not retransmitted by the receiver after being lost.

D. Frame 2 is not retransmitted by the sender.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Consider a sliding window flow control protocol between hosts A and B. Suppose the link propagation delay is 1 time unit, the retransmission timeout is 3 time units, and the window size is 3. Assume the link drops every third data packet, i.e., the link drops the 1 st, $4th, 7th, \ldots$. data packets. (Note that here "kth packet" means the kth packet transmitted on the link, and not the sequence number of the packet.) How long (in ms) does it take to transmit 6 packets between A and B ? (Note: Ignore the transmission times and the queueing delay, and assume that no acknowledgements are lost.) Assume sender has an unlimited number of packets to send.

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff